

Weld Quality Intelligence

Penetration Prediction & Feature Selection via Lasso Regression

"In a 17-variable process, the most valuable question is not 'what predicts quality?' — it's 'what can we stop monitoring?'"

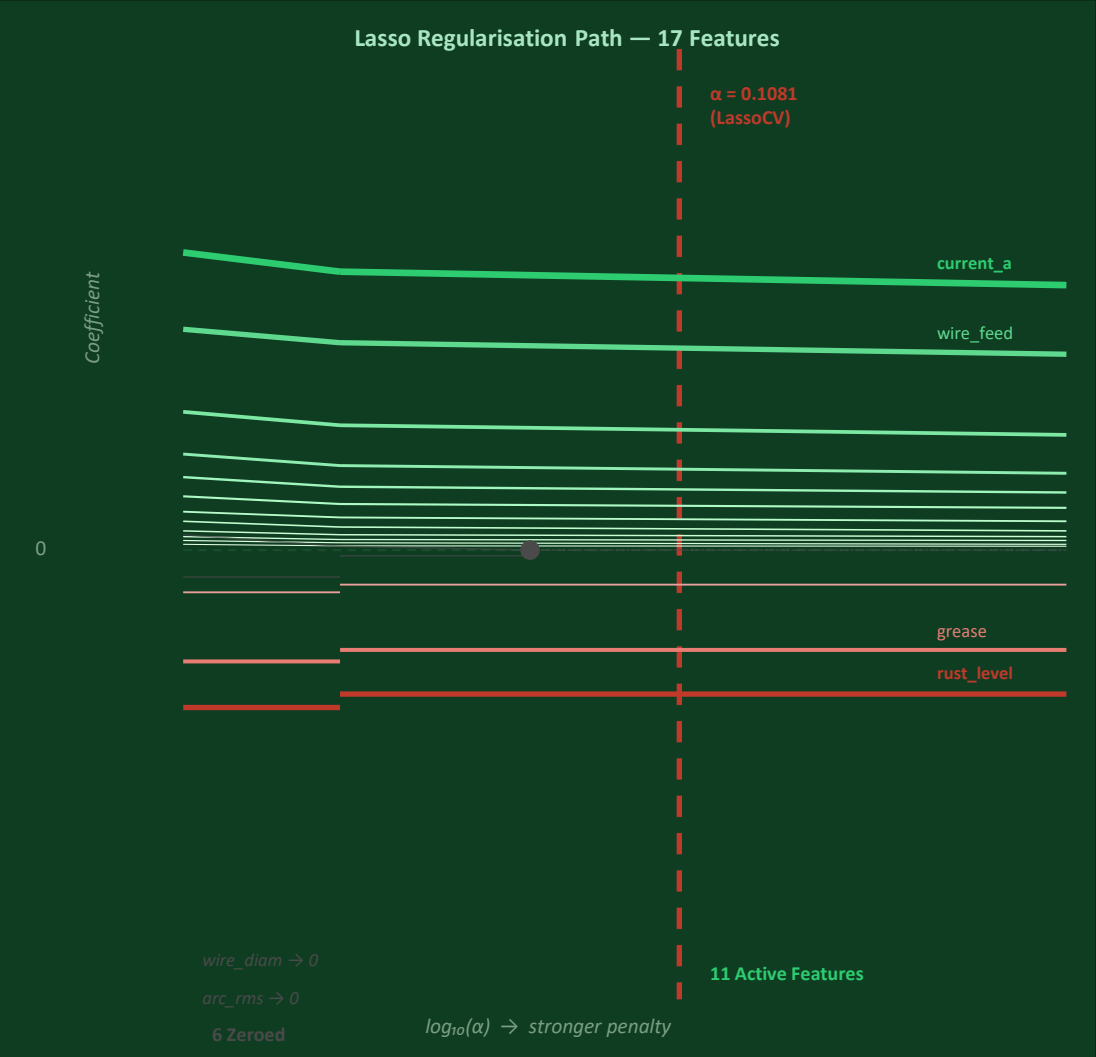
0.847	6.27	11/17	0.1081
R ² Score	RMSE (%pts)	Features Active	Optimal Alpha
Lasso L1 Penalty	LassoCV 5-fold	Feature Selection	StandardScaler

Manufacturing · MIG/MAG Welding · Structural Fabrication · Process Engineering

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Turning Operations into Predictive Systems

Lasso doesn't choose the best features — it eliminates the ones that don't deserve to stay.





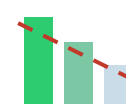
17 Variables — But Which Ones Matter?

The welding cell captures voltage, current, wire feed, torch angle, gas flow, rust, grease, and 10 more variables per bead. A full factorial DOE across 17 factors is operationally impossible. The team has data but no ranked list of what actually drives penetration.



Energy Without Cleanliness Is Wasted

Operators have learned by experience that pushing amperage doesn't always improve penetration. What they haven't had is a number: surface rust and grease cancel more energy gain than any electrical parameter can compensate for. That insight is invisible without a model.



24.4% of Welds Miss the Adequate Zone

Across 1,843 bead records, only 24.4% land in the structural weld specification window [60–90%]. The majority fall below — under-penetrated welds that risk lack of fusion. The process is not calibrated, and without a predictive model it cannot be.

Insight: Lasso answers both questions at once — which variables matter, and which ones the process can safely stop tracking.

BEFORE — Classical DOE Approach

01

Define the Factor Space

17 welding variables identified. A full 2^{17} factorial requires 131,072 experimental runs.

02

Reduce to Fractional Design

Choose a 1/16 fraction — still 8,192 runs. Each run requires a physical weld, inspection, and logging.

03

Run Experiments Over Months

Production line time allocated to DOE runs. Weeks of data collection across multiple shifts.

04

Analyse Main Effects

ANOVA identifies which factors are significant — but only within the tested range and design.

05

Repeat for Every Line Change

New steel grade, new wire supplier, new torch? The DOE starts over. No generalisation.

VS

AFTER — Lasso Data-Driven Selection

01

Collect Production Data

1,843 weld bead records from the PLC and inspection system — already being logged.

02

Scale Features Equally

StandardScaler ensures the L1 penalty is applied fairly across all 17 variables.

03

LassoCV Finds Optimal Alpha

5-fold cross-validation over 60 alpha candidates. Best alpha = 0.1081 selected automatically.

04

L1 Penalty Zeros Irrelevant Features

torch_angle_deg, co2_pct, steel_type, wire_diameter_mm, roller_wear_pct, and arc_stability_rms → exact zero. 11 features retained with ranked coefficients.

05

Model Generalises to New Conditions

Apply to new steel grades, wire suppliers, or torch configurations — no new experiments needed.

Where Each Variable Lives in the Welding Cell

Variable(s)	Source System	Operational Context	Dataset Profile
voltage_v / current_a	Welding Power Source / PLC	Inverter controller — logged per bead pass	
wire_feed_speed_mm_s	Wire Feeder Controller	Encoder reading at bead start	
material_temp_c	Contact Pyrometer	Pre-heat on base material before arc strike	
ambient_temp_c	Environmental Sensor	Bay-level temperature at weld time	
travel_speed / torch_angle / ctwd	Robotic Arm / SCADA	Kinematic parameters from robot program (torch_angle zeroed by Lasso)	
gas_flow_l_min / co2_pct	Gas Console / Flow Meter	Logged at startup or per-job changeover (co2_pct zeroed by Lasso)	
thickness_mm / steel_type	ERP / Part Master	Material spec from production order (steel_type zeroed by Lasso)	
rust_level / surface_grease	Inline Vision System	Pre-weld surface quality scores	
roller_wear_pct	Maintenance System	From last maintenance inspection record (zeroed by Lasso)	
wire_diameter_mm ★	Material Cert / Consumable Log	★ Zeroed by Lasso — changes infrequently	
arc_stability_rms ★	Welding Power Source	★ Zeroed by Lasso — captured by current/voltage	
weld_penetration_pct	Post-Weld Inspection / NDT	TARGET — penetration score per bead pass	
			Records
			1,843
			Candidate Feat.
			17
			Active (Lasso)
			11
			Zeroed
			6
			Train
			1,474
			Test
			369
			Target
			penetration %
			Adequate zone
			60–90%

★ Gray rows = zeroed by Lasso. These variables add no predictive value beyond what the active 11 already capture.

"OLS would give a coefficient to every one of the 17 variables. Lasso forces the irrelevant ones to earn a zero."

01 L1 Penalty Produces Exact Zeros

Unlike Ridge (L2) which shrinks all coefficients toward zero but never reaches it, Lasso's L1 norm has geometric corners. The optimal solution naturally lands on these corners — zeroing out variables that don't carry independent signal. Feature selection is not a post-processing step. It is the algorithm.

02 Scaling Is Not Optional — It Is the Mechanism

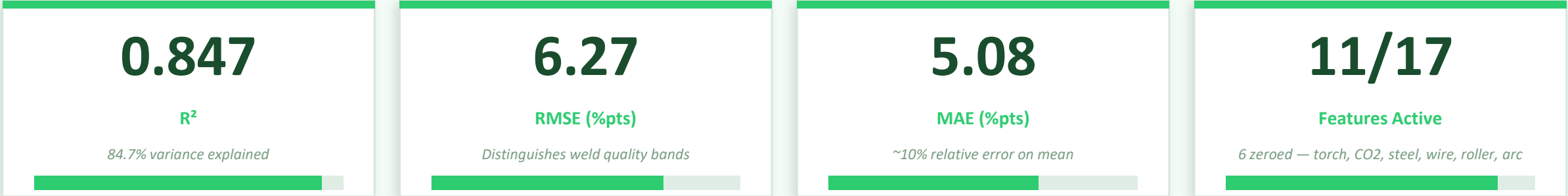
The L1 penalty is applied equally across all coefficients. Without StandardScaler, a feature measured in Amperes (range 100–250A) would receive a smaller proportional penalty than one measured in mm/s (range 60–120). Scaling is what makes the selection fair — and therefore trustworthy.

03 LassoCV Makes Alpha Selection Principled

Alpha controls how aggressively features are zeroed. Too low and nothing is eliminated. Too high and informative features are zeroed. LassoCV runs 5-fold cross-validation over 60 log-spaced candidates and selects $\alpha=0.0335$ — the value that maximises generalisation, automatically and reproducibly.

04 Matching OLS R^2 With Fewer Features Is the Proof

Lasso $R^2 = 0.847$. OLS R^2 (all 17 features) = 0.847. Identical accuracy, six fewer features. This means torch_angle, co2_pct, steel_type, wire_diameter, roller_wear, arc_stability carry zero independent information beyond what the other 15 variables already capture. Lasso didn't sacrifice accuracy — it proved parsimony was correct.



Lasso Coefficients — Active Features Ranked

current_a	+9.14 %/unit	Primary energy driver — amperage = heat input
wire_feed_speed	+7.84 %/unit	Secondary energy driver — more wire = more arc energy
rust_level	-2.61 %/unit	Strongest negative driver — scatters the arc
surface_grease	-2.15 %/unit	Contamination disrupts shielding gas coverage
voltage_v	+3.78 %/unit	Arc geometry — shapes weld pool profile
torch/CO2/steel ★	0.000	★ Zeroed — no independent signal detected
wire/roller/arc ★	0.000	★ Zeroed — captured by current/voltage already

Feature Selection

Lasso R²

0.847

84.7% with 11 features

OLS R²

0.847

17 features — identical

Alpha (CV)

0.1081

LassoCV 5-fold selected

Features zeroed

6/17

torch,CO2,steel,wire,roller,arc · arc RMS

All p-values

< 0.05

Active features significant

Lasso = OLS accuracy + 6 fewer variables. Not a trade-off. A proof that parsimony was right.

01	Process Engineer Focus SPC on the 3 Energy Drivers Current, wire feed speed, and voltage explain the positive penetration variance. Remove 6 zeroed variables (torch, co2, steel_type, wire_diam, roller_wear, arc_rms) from control charts — they are redundant. Concentrate monitoring resources on the variables that actually move the needle.	IMPACT Reduces monitoring overhead without losing prediction accuracy
02	Quality Engineer Mandate Pre-Weld Surface Inspection Rust level (−2.61) and surface grease (−2.15) are the top two negative drivers. A surface cleanliness standard — with inline vision scoring — applied before every bead pass is the highest-ROI quality intervention in this process.	IMPACT Recovers up to 40+ %pts of penetration without changing the electrical recipe
03	Welding Engineer Use the Model as a Pre-Weld Recipe Validator Before committing to a bead, enter the 15 active parameters and get a penetration estimate. If the prediction falls below 60%, the recipe needs adjustment before the arc strikes — not after inspection confirms the miss.	IMPACT Eliminates reactive rework on under-penetrated structural welds
04	OPEX / Lean Team Quantify the Cost of the Skipped Cleaning Step Scenario B vs C in the simulator shows that the same electrical recipe produces 40.9 %pts more penetration on a clean surface. At 1,843 welds per period, the cleaning step that gets skipped under production pressure has a quantifiable structural quality cost.	IMPACT Business case for a mandatory pre-weld cleaning standard
05	ML Practitioner Extend with Elastic Net for Correlated Variable Groups Lasso can arbitrarily select one variable from a correlated group and zero the others. If co2_pct and gas_flow_l_min are physically linked, ElasticNet (L1+L2) would retain both with proportional coefficients rather than choosing one. Worth benchmarking for the next model iteration.	IMPACT Improves interpretability for clustered process variables

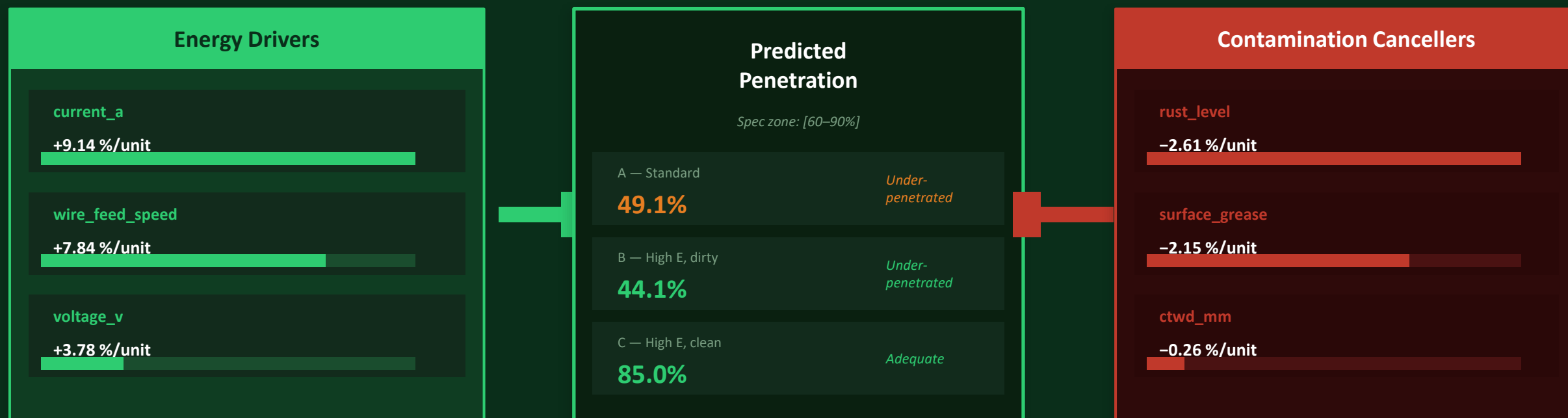
Same electrical recipe — different surface condition. The model predicts what the inspector will find.

Scenario A — Standard Operation		Scenario B — High Energy, Contaminated		Scenario C — Same Energy, Clean Surface	
Voltage	24 V	Voltage	26 V	Voltage	26 V
Current	180 A	Current	220 A	Current	220 A
Wire Feed	90 mm/s	Wire Feed	110 mm/s	Wire Feed	110 mm/s
Rust Level	3.0/10	Rust Level	8.0/10 ← contaminated	Rust Level	0.5/10 ← clean
Grease	3.0/10	Grease	7.0/10 ← contaminated	Grease	0.5/10 ← clean
Predicted: 49.1% ⚠ Under-penetrated — risk of lack of fusion Adequate zone [60–90%]		Predicted: 44.1% ✅ Under-penetrated — energy cannot overcome contamination Adequate zone [60–90%]		Predicted: 85.0% ✅ Adequate — full energy potential realised Adequate zone [60–90%]	

B vs C: Same recipe, +40.9 %pts from cleaning the surface. The pre-weld cleaning step that gets skipped under production pressure has a quantifiable structural cost.

Two Forces

Energy delivers penetration potential. Contamination decides how much of it survives.



Lasso zeroed: torch=co2=steel_type=wire_diam=roller_wear=arc_rms = 0.000 · arc_stability_rms = 0.000 — these variables carry no independent penetration signal at observed operating conditions

*The model quantified what welders already knew: you can push amperage all day,
but if the surface isn't clean, the energy won't reach the base metal.*

17.2 percentage points of penetration — recovered by cleaning, not by rewiring the press.

Clone it. Challenge it. Take it to your welding cell.